

PACHYCEREUS

CACTACEAE

Genus of slow-growing, columnar, perennial cacti, branching with age. The funnel-shaped flowers are unlikely to be borne in cultivation, as they develop only on plants over 10ft (3m) high. Frost tender. Requires sun and well-drained soil. Propagate by seed in spring or summer. ***P. marginatus***, syn. *Marginatocereus marginatus*, *Stenocereus marginatus* (Organ-pipe cactus), illus. p.494. ***P. pecten-aboriginum***. Columnar, perennial cactus. H 25ft (8m), S 10ft (3m). Dark green stems bear 10–11 deep ribs. Each areole has 8–9 radial spines, ½in (1cm) long, and 1–2 longer central spines. Red-based, dark brown spines fade to gray. In summer, bears night- and day-blooming, funnel-shaped white flowers, with greenish-red outer petals. Z11–12 (min. 50°F/10°C) ***P. pringlei*** illus. p.483. ***P. schottii***, syn. *Lophocereus schottii*, illus. p.493. ‘**Monstrosus**’ is a columnar, perennial cactus. H 10ft (3m), S 3ft (1m). Irregular, olive- to dark green stems have 4–9 ribs and no spines. Produces funnel-shaped, pink, red, or white flowers, 1¼in (3cm) across, at night in summer. Z11–12 (min. 50°F/10°C)

PACHYPHRAGMA

BRASSICACEAE/CRUCIFERAE

Genus of one species of semievergreen, rhizomatous perennial, grown for groundcover and flowers. Fully hardy. Needs moist, fertile, leafy soil in partial shade; good for planting beneath trees and deciduous shrubs. Propagate by seed in fall or by division in spring. ***P. macrophyllum***, syn. *Thlaspi macrophyllum*, illus. p.255.

PACHYPHYTUM

CRASSULACEAE

Genus of rosetted, perennial succulents, closely related to *Echeveria*, with which it hybridizes. Frost tender. Needs sun and well-drained soil. Propagate by seed, or leaf or stem cuttings in spring or summer. ***P. compactum*** illus. p.491. ***P. oviferum*** (Moonstones, Sugar almond-plum) illus. p.490.

PACHYPODIUM

APOCYNACEAE

Genus of bushy or treelike, perennial succulents, mostly with swollen stems, closely related to *Adenium*, except that most species have spines. Frost tender. Requires full sun and very well-drained soil. May be very difficult to grow. Propagate by seed in spring or summer. ***P. lamerei*** illus. p.494. ***P. succulentum***. Treelike, perennial succulent. H 2ft (60cm), S 1ft (30cm). Swollen trunk has narrow, green to gray-brown stems with paired thorns. Bears

silverform, pink, white, sometimes red-striped or red, flowers near the stem tips in summer. Z12 (min. 59°F/15°C)

PACHYSANDRA

BUXACEAE

Genus of evergreen or semievergreen, creeping perennials and subshrubs, grown for their tufted foliage. Useful for ground-cover. Fully hardy. Tolerates dense shade and grows in any but very dry soil. Propagate by division in spring. ***P. terminalis*** illus. p.375. ‘**Green Sheen**’, H 8in (20cm), S indefinite, is a compact, slow-growing, evergreen shrublet. Has clusters of striking, glossy, coarsely toothed, dark green leaves, at the ends of short stems. Bears tiny, white flowers in small spikes, in early summer. Z5–9. ‘**Variegata**’, H 10in (25cm), S to 2ft (60cm), is spreading, with diamond-shaped, cream-variegated leaves. Flowers are sometimes flushed purple.

PACHYSTACHYS

ACANTHACEAE

Genus of evergreen perennials and shrubs, grown for their flowers. Frost tender. Needs partial shade and fertile, well-drained soil. Water containerized plants freely when in full growth, moderately at other times. Cut back flowered stems in late winter to maintain a bushy habit. Propagate by greenwood cuttings in early summer. Whitefly and red spider mite may cause problems. ***P. cardinalis***. See *P. coccinea*. ***P. coccinea***, syn. *Jacobinia coccinea*, *Justicia coccinea*, *P. cardinalis* (Cardinal’s guard). Evergreen, erect, robust shrub. H to 6ft (2m), S 2–3ft (60cm–1m). Leaves are ovate and deep green. Bears tubular, bright red flowers in tight, green-bracted spikes, 6in (15cm) long, in winter. Z11–12 (min. 55°F/13°C; min. 59–64°F/15–18°C to flower well) ***P. lutea*** (Lollipop plant) illus. p.459.

X PACHYVERIA

CRASSULACEAE

Hybrid genus (*Echeveria* × *Pachyphytum*) of clump-forming, rosetted, perennial succulents, sometimes almost stemless. Frost tender. Requires full sun or partial shade and very well-drained soil. Propagate by leaf or stem cuttings in spring or summer. **× *P. glauca***, syn. *Echeveria* × *fruticosa*, illus. p.490.

PAEONIA

Peony

PAEONIACEAE

Genus of late spring-flowering perennials and deciduous shrubs or subshrubs (“tree peonies”), valued for their bold foliage, showy blooms, and, in some species, colorful seed pods. Fully to frost hardy; most are fully hardy, although young growth (especially on tree peonies) may be damaged by late spring frosts. Prefers sun (but tolerates light shade) and rich, well-drained soil. Tall and very large-flowered cultivars need support. Propagate all species by seed in fall (may take up to

3 years to germinate), tuberous species by root cuttings in winter, tree peonies by semiripe cuttings in late summer, or by grafting in winter. Perennials may also be propagated by division in fall or early spring. Prone to peony wilt. ☹All parts can cause mild stomach upset if ingested. See also feature panel pp.228–229.

Flower forms

Unless stated otherwise, peonies described below flower between late spring and early to midsummer and have large, alternate leaves divided into ovate to lance-shaped or linear leaflets. Flowers are single, semidouble, double, or anemone-form.

Single—flowers are mostly cup-shaped, with 1 or 2 rows of large, often lightly ruffled, incurving petals and a conspicuous central boss of stamens.

Semidouble—flowers are similar to single ones, but have 2 or 3 rows of petals.

Double—flowers are rounded, usually composed of 1 or 2 outer rows of large, often lightly ruffled, incurving petals, the remaining petals being smaller, usually becoming more densely arranged and diminishing in size toward the center.

Stamens are few, inconspicuous, or absent.

Anemone-form (Imperial or Japanese)—flowers usually have 1 or 2 rows of broad, incurving, outer petals; the center of the flower is often filled entirely with numerous densely arranged, sometimes deeply cut, narrow petaloids derived from stamens.

P. ‘Alice Harding’. See *P. × lemoinei* ‘Alice Harding’. ***P. anomala* subsp. *veitchii***, syn. *P. veitchii* (illus. p.229). Clump-forming perennial. H and S to 30in (75cm). Has shiny, divided, bright green leaves and produces nodding, cup-shaped, single, purple-pink flowers in late spring and early summer. Z7–10 ***P. ‘Argosy’***. See *P. × lemoinei* ‘Argosy’. ***P. arietina***, syn. *P. mascula* subsp. *arietina*. Tuberous perennial. H and S to 30in (75cm). Dark green foliage, often with lobed leaflets, is paler and hairy beneath; stems are dark red. Bears single, cup-shaped, reddish-pink flowers in late spring. Seed capsules of 2–5 boat-shaped sections split to reveal purplish-black seeds. Z7–10 ***P. ‘Avant Garde’***. Clump-forming perennial. H and S to 3ft (1m). Has luxuriant foliage. Medium-sized to large, fragrant, single flowers are pale rose-pink with darker veins and bright golden anthers that have yellow-red filaments. Flowers are borne on stiff, straight stems in mid-spring and are ideal for cutting. Z7–10 ***P. ‘Bartzella’*** (Itoh hybrid; illus. p.229). Clump-forming, free-flowering hybrid between a herbaceous perennial and a tree peony. H and S 3ft (90cm). In early summer, produces large, scented, double, lemon-yellow flowers, marked with red. Z7–10 ***P. ‘Bowl of Beauty’*** (illus. p.228). Clump-forming perennial. H and S to 3ft (1m). Bears very large, striking, anemone-form flowers with pale carmine-pink, outer petals and numerous, narrow, densely arranged, ivory-white, inner petaloids. Z7–10 ***P. ‘Buckeye Belle’*** (illus. p.229) Clump-forming perennial. H and S 3ft (90cm). Has red-tinged, mid-green leaves and bears semidouble, dark red flowers with a mass of central, golden stamens. Z7–10

P. cambessedesii (Majorcan peony; illus. p.228). Clump-forming perennial. H and S 1½ft (45cm). Has especially attractive foliage, dark green above with veins, stalks and undersurfaces suffused purple-red. Single, deep rose-pink flowers are borne in mid-spring. Z9–11 ***P. ‘Cheddar Cheese’***. See *P. lactiflora* ‘Cheddar Cheese’. ***P. ‘Cheddar Gold’***. See *P. lactiflora* ‘Cheddar Gold’. ***P. ‘Chocolate Soldier’***. Clump-forming perennial. H and S to 3ft (1m). Has mid- to dark green leaves that are often tinged bronze-red when young. Semidouble, deep purple-red flowers, with golden stamens, are borne in early summer. Z7–10 ***P. ‘Claire de Lune’*** (illus. p.229). Clump-forming perennial. H and S 28–32in (70–80cm). In mid-spring, single, pale yellow flowers, each with a central boss of golden stamens, are borne in pairs. Z7–10 ***P. ‘Coral Charm’*** (illus. p.228). Clump-forming perennial. H and S 28in (70cm). Has large, divided, dark green leaves. Bears semidouble, peach-pink flowers, opening from darker buds, in late spring or early summer. Z7–10 ***P. ‘Coral Fay’***. Clump-forming perennial forming a rather spreading dome. H and S 32–36in (80–90cm). Semidouble, pink-tinged, rich crimson flowers are paler toward the petal bases; stamens are golden-yellow. Z7–10 ***P. corallina***. See *P. mascula* subsp. *mascula*. ***P. ‘Cora Louise’*** (Itoh hybrid). Clump-forming, free-flowering hybrid of perennial and tree peony. H 24–32in (60–80cm), S 2ft (60cm). Bears large, scented, semidouble, white flowers with petals marked purplish at the bases. Z7–10 ***P. ‘Cornelia Shaylor’***. See *P. lactiflora* ‘Cornelia Shaylor’. ***P. daurica* subsp. *mlokosewitschii***. See *P. mlokosewitschii*. **subsp. *wittmanniana*** see *P. wittmanniana*. ***P. decora***. See *P. peregrina*. ***P. ‘Defender’***. Clump-forming, vigorous perennial. H and S to 3ft (1m). Single, satiny crimson flowers, to 6in (15cm) across, with a central boss of golden anthers, are borne on strong stems. Z7–10 ***P. delavayi*** (Tree peony) illus. p.136. ***var. angustiloba*** is a deciduous, upright, open, suckering shrub or subshrub (tree peony). H and S to 3ft (1m). Leaves are deeply divided into pointed-ovate leaflets, often with reddish stalks. In late spring, bears small, bowl-shaped, red, red-purple, yellow, orange, or white flowers, with conspicuous, leafy bracts beneath. Z7–10. ***var. angustiloba f. alba*** (syn. *P. potaninii f. alba*) has white flowers. ***var. angustiloba f. angustifolia*** (syn. *P. potaninii*) produces dark red, red, or reddish-purple flowers. ***var. angustiloba f. trollioides*** (syn. *P. potaninii* var. *trollioides*, *P. trollioides*) has yellow or orange flowers. ***var. delavayi f. delavayi*** has dark red to purplish flowers. ***var. delavayi f. lutea*** (syn. *P. lutea*; illus. p.229) has orange, yellow, or greenish-yellow flowers, sometimes red at the bases or on the petal margins. ***var. ludlowii*** see *P. ludlowii*. ***P. ‘Early Windflower’***. Clump-forming perennial. H 28in (70cm), S 20in (50cm).